

Kevin Kattakayam

Backend & Distributed Systems Engineer | Go, Rust, Kafka, Kubernetes | Internship & Full-Time | Graduating 2027

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EDUCATION

Karunya Institute of Technology and Sciences

2023 – 2027 (Expected)

B.Tech, Computer Science and Engineering (AI & ML) – CGPA: 8.39/10

Coimbatore, India

Viswajyothi CMI Public School, Angamaly – Secondary, CBSE (2021): 92.2%; Higher Secondary, CBSE (2023): 86%

EXPERIENCE

TheAiSignal – Full Stack AI Developer Intern

May 2026 – Jul 2026

- Built and shipped FastAPI backend services and REST API layers powering production application features.
- Maintained deployment pipelines and release workflows across multiple projects using GitHub and Cloudflare.

Karunya Computer Technology Centre – Generative AI Trainee

Dec 2024 – Present

- Build LLM-powered applications with a team across multiple generative AI projects using LangChain, ChromaDB, and FAISS.
- Apply retrieval-augmented generation, prompt engineering, and OCR-based extraction to document processing workflows.

YoungMinds Technology Solutions Pvt Ltd – Generative AI Intern

May 2025 – Jul 2025

- Developed automation tools in Python against LLM and REST APIs.

Karunya Innovation and Design Studio (KIDS) – Generative AI Intern

May 2025 – Jun 2025

- Built a multi-agent patent drafting system in CrewAI, orchestrating 5 agents for prior-art search, claim generation, and IPO compliance checking.

Octanet Services Pvt Ltd – Web Development Intern

May 2024 – Jun 2024

- Built responsive web applications using HTML, CSS, and JavaScript following mobile-first design principles.

PROJECTS

Distributed Observability Pipeline (Datadog-style)

github.com/KevinKattakayam/datadog

Go, Rust, Apache Kafka, ClickHouse, Prometheus, Grafana, Kubernetes, Terraform

- Built an end-to-end metrics pipeline: a Go HTTP ingestor (Gin) publishing validated metrics to 6-partition Kafka topics with a dead-letter queue, consumed by a Rust/tokio processor running EWMA and rolling Z-score anomaly detection.
- Ensured at-least-once delivery by committing Kafka offsets only after successful ClickHouse writes, and added a circuit breaker (opens after 5 failures, backoff to 60s) so backend degradation bounds consumer lag.
- Built the operational layer: per-tenant rate limiting, W3C Trace Context propagation to Jaeger, multi-window SLO burn-rate alerts, a 12-component Helm chart with HPAs and PDBs, and a Terraform blueprint for AWS EKS.
- Backed it with 53 unit tests, 25 integration tests, and a k6 load-test script for the 14-container stack.

DevSync – Collaborative Developer Workspace

github.com/KevinKattakayam/devsync

Node.js, Express, TypeScript, PostgreSQL, Prisma, Socket.io, Next.js

- Built a real-time collaboration backend: JWT auth, RBAC, Socket.io document co-editing, and Kanban tasks over Prisma/PostgreSQL.
- Added Zod validation, centralized error handling, and Jest/Supertest suites across auth, RBAC, and task workflows.

PharmaTrace – Pharmaceutical Verification Platform

github.com/KevinKattakayam/pharma_trace

Python, FastAPI, LangGraph, Pydantic, Supabase PostgreSQL/PostGIS

- Built a FastAPI service running a 6-agent LangGraph pipeline over OpenFDA, RxNav, and CDSCO with parallel `asyncio.gather` lookups.
- Enforced strict Pydantic schemas, logged verifications to a SHA-256 hash-chained audit log, and stored geospatial data in PostGIS.

NexusBank – Online Banking API

github.com/KevinKattakayam/online_bank_simulation

Node.js, Express 5, MongoDB, Mongoose, bcrypt, express-session

- Built a transactional banking API with bcrypt-hashed PIN auth, MongoDB-backed sessions, balance-validated transfers, compound-indexed paginated history, and rule-based fraud scoring over aggregation pipelines.

TECHNICAL SKILLS

Languages: Go, Rust, Python, TypeScript, JavaScript, Java, SQL

Backend: FastAPI, Node.js, Express, Gin, gRPC, REST APIs, Prisma, Mongoose, Socket.io, JWT, RBAC, Pydantic, Zod

Distributed Systems: Apache Kafka (KRaft), Schema Registry, Kafka Connect, at-least-once delivery, circuit breakers, rate limiting

Data Stores: PostgreSQL, ClickHouse, MongoDB, Redis, Snowflake, ChromaDB, FAISS

Infra & Observability: Docker, Kubernetes, Helm, Terraform, Prometheus, Grafana, OpenTelemetry, Jaeger, CI/CD, Cloudflare, k6

Foundations: Data Structures & Algorithms, System Design, Concurrency, Distributed Tracing, SLOs

ACHIEVEMENTS & CERTIFICATIONS

- IEEE Publication (2026):** “Early Lung Cancer and Heart Disease Prediction from Clubbed Fingers Using Machine Vision”.
ieeexplore.ieee.org/document/11469562
- Completed Amazon ML Summer School 2026** (Amazon)

Certifications: SnowPro Associate: Platform (Snowflake) · Redis Associate Developer · SQL AI Developer Associate, DP-800 (Microsoft) · Azure AI Engineer Associate, AI-102 (Microsoft) · Google Cybersecurity Professional · LLM Security Professional, CLLMSP